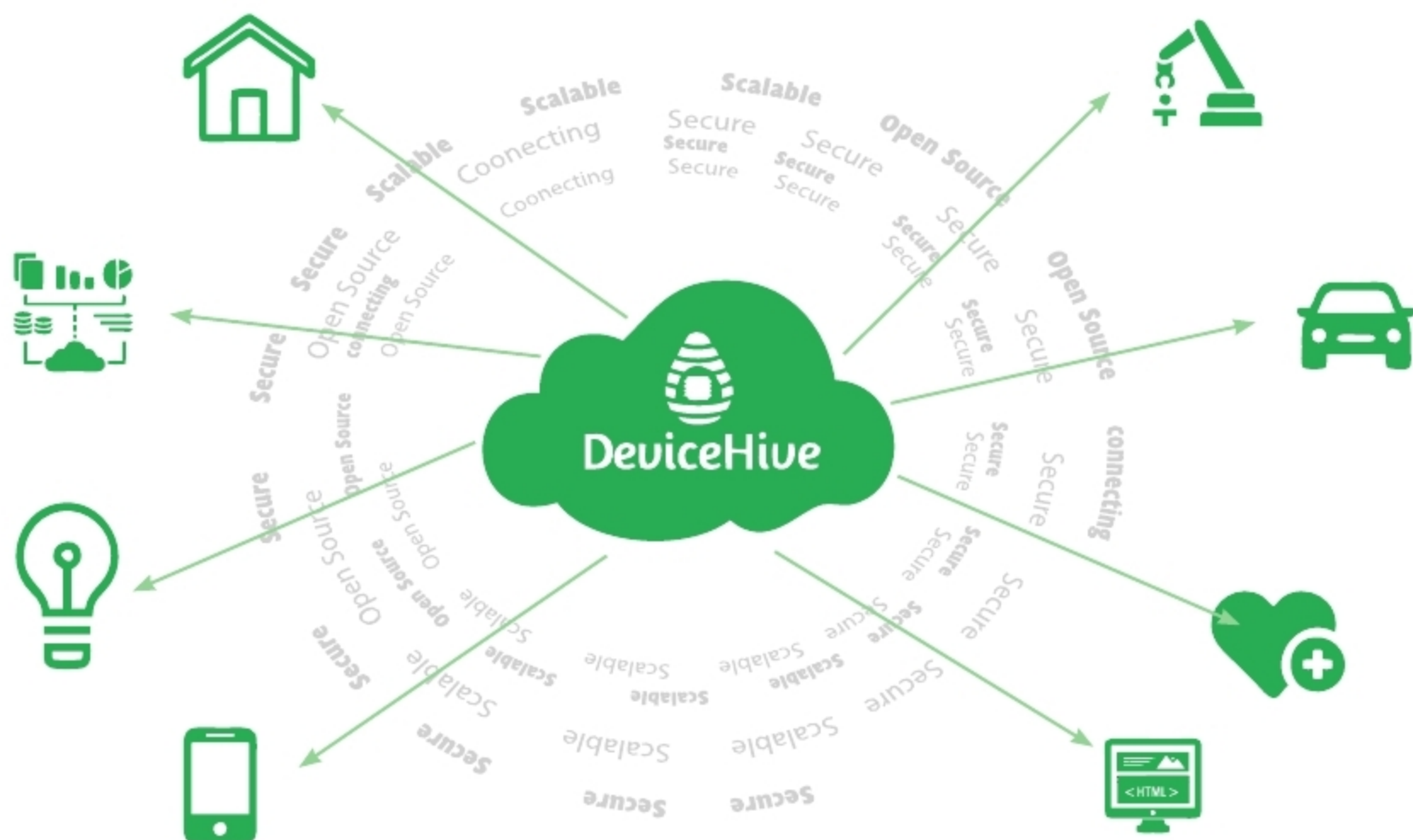


DeviceHive: The Scalable Open Source M2M Development Platform

DeviceHive provides powerful instruments for smart devices to communicate and manage services. It incorporates three critical technologies that affect mobile developers and users— the cloud, mobile and embedded systems.

It consists of a communication layer, control software and multi-platform libraries and clients to bootstrap the development of remote sensing, remote control, monitoring and automation, smart energy, etc.



Today, people look for easy ways to get things done and the best example of this is automation. Machine-to-machine (M2M) communication aims to connect everyday objects and allows them (these non-human content providers) to feed the Internet with data in various formats, frequently. M2M communication is the latest trend in the evolution of industry, combining technology with data communication between devices or machines.

M2M technology was first implemented in the manufacturing and industrial sectors, where other technologies like SCADA and remote monitoring helped to remotely manage

and control data from equipment. M2M communications is all about direct inter-device communications, through which a robot or machine controls other machines. It can be used to more effectively monitor the condition of critical public infrastructure such as water treatment facilities or bridges, with less human intervention.

Making a machine-to-machine communications system work is a step by step procedure. The three main elements in this process are: sensors (that acquire data from the operational environment and transmit it wirelessly), peer-to-peer wireless networks, and Internet enabled PCs. The most common types of M2M

communications are listed below.

- **Backend-to-backend:** This is all about transmitting device logs over the Internet to the cloud provider. It also works for schedulers, daemons and continuous processes.
- **IoT devices:** These small connected units put together data from small, autonomous, specialised devices at the server.
- **CLI clients:** This is the creation of CLI apps that have the necessary rights to perform the actions, but which are only available on certain computers.

The following points highlight the architecture and components of M2M communication.